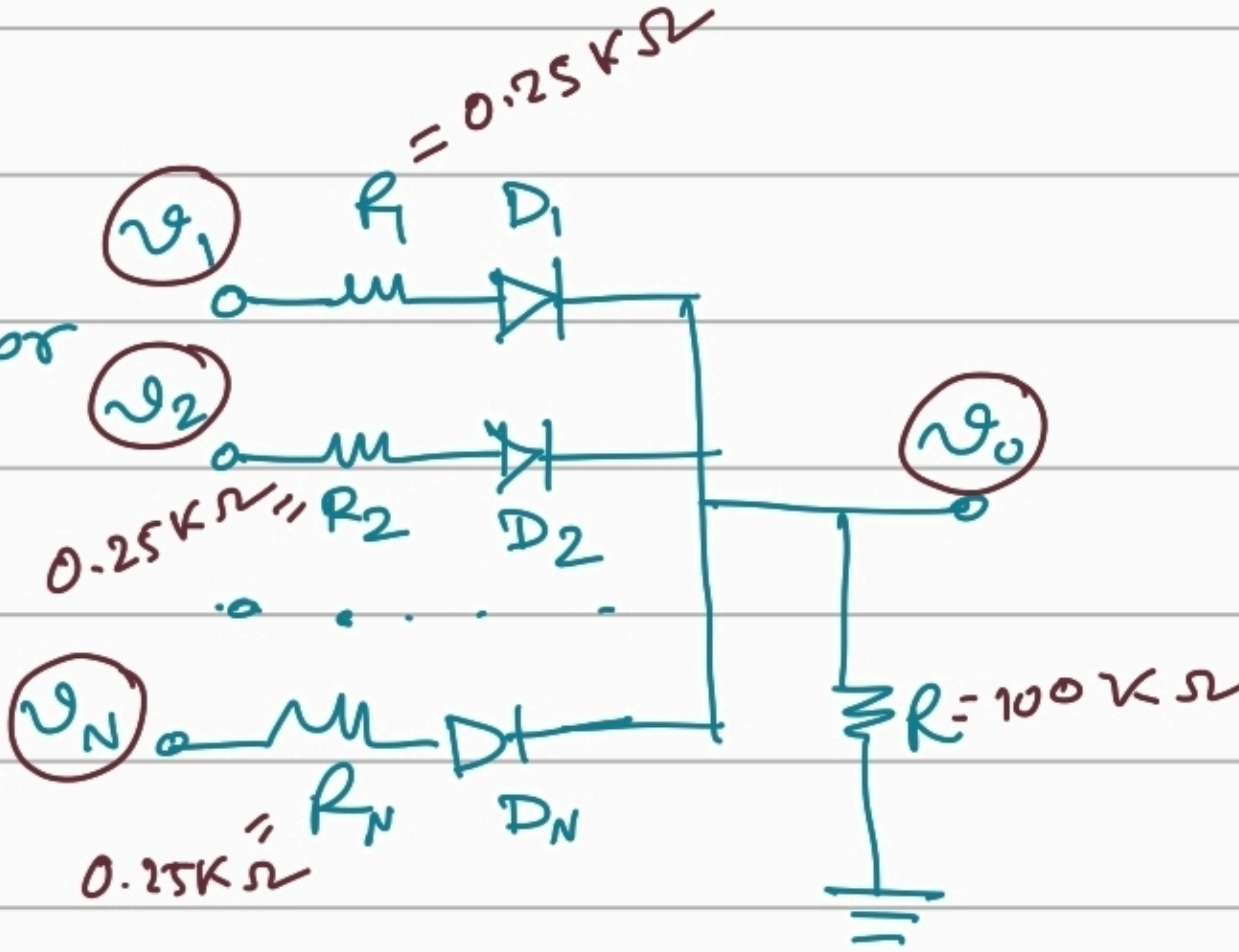
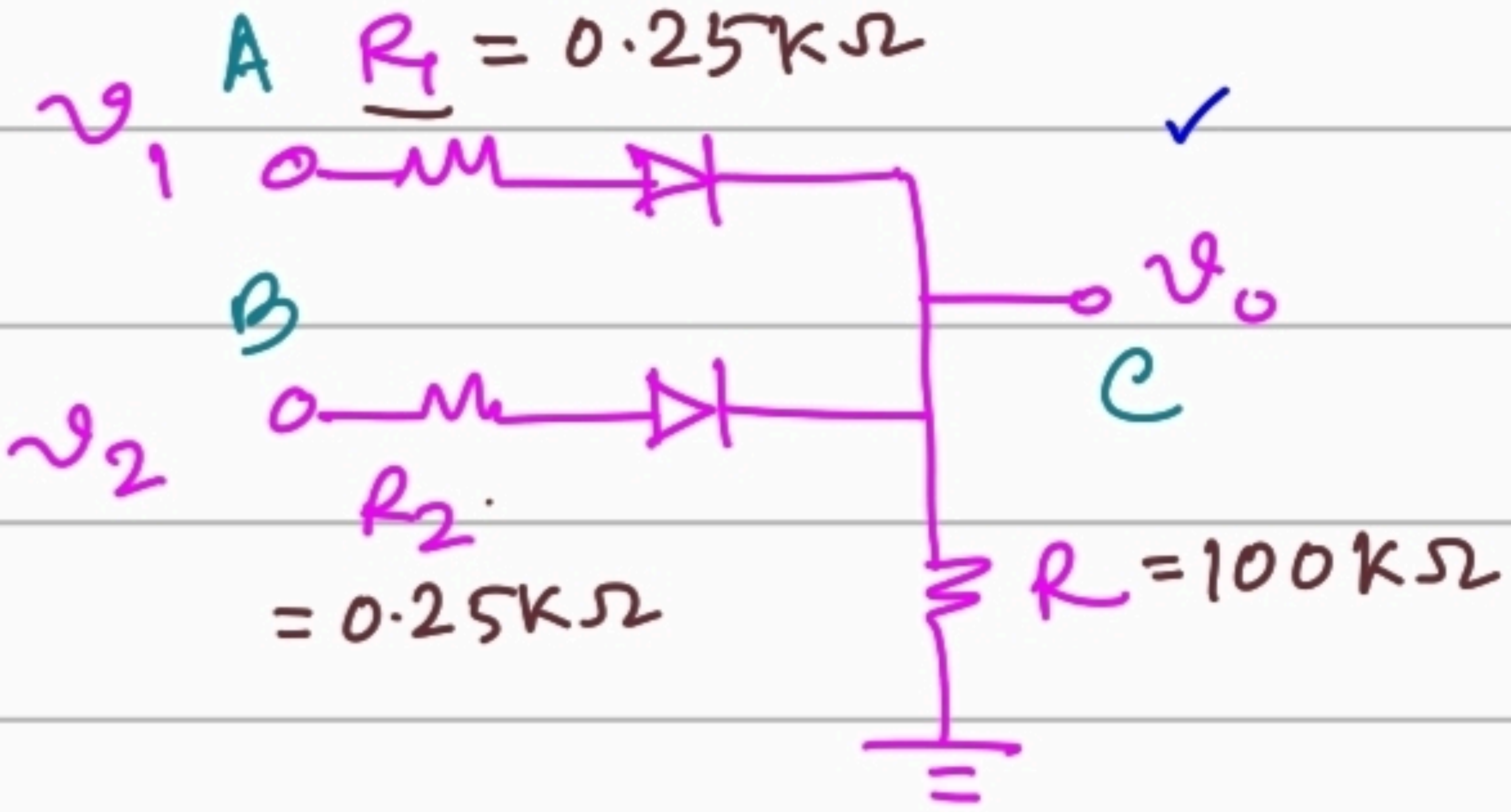
Diode Logic :-

OR GATE :-

N number of inputs for OR gate



Operation : 2-OR GATE



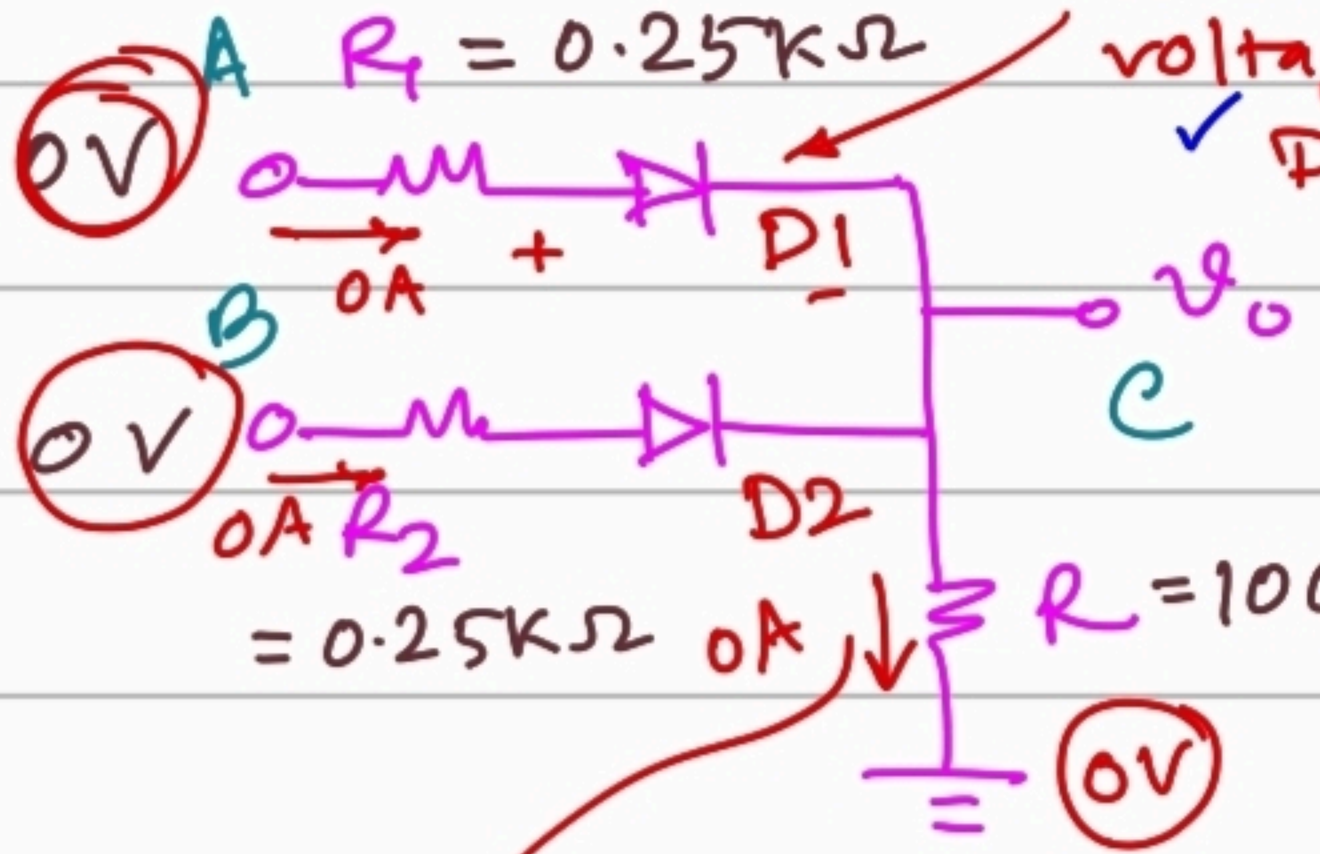
	A	B	v_1	v_2	v_o	C
1	0	0	0V	0V	0V	0
2	0	1	0V	5V	4.289	1
3	1	0	5V	0V	4.289	1
4	1	1	5V	5V	4.294	1

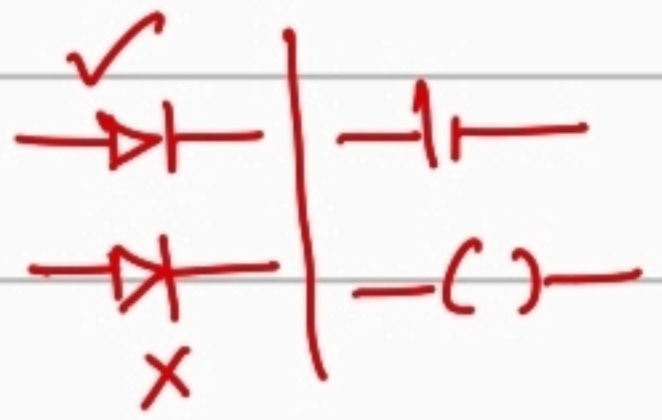
We will assume logic 1 for input part is 5V.
and logic 0 for input part is 0V.

Case 1:

$v_1 = 0V, \; v_2 = 0V$

constant voltage Drop.

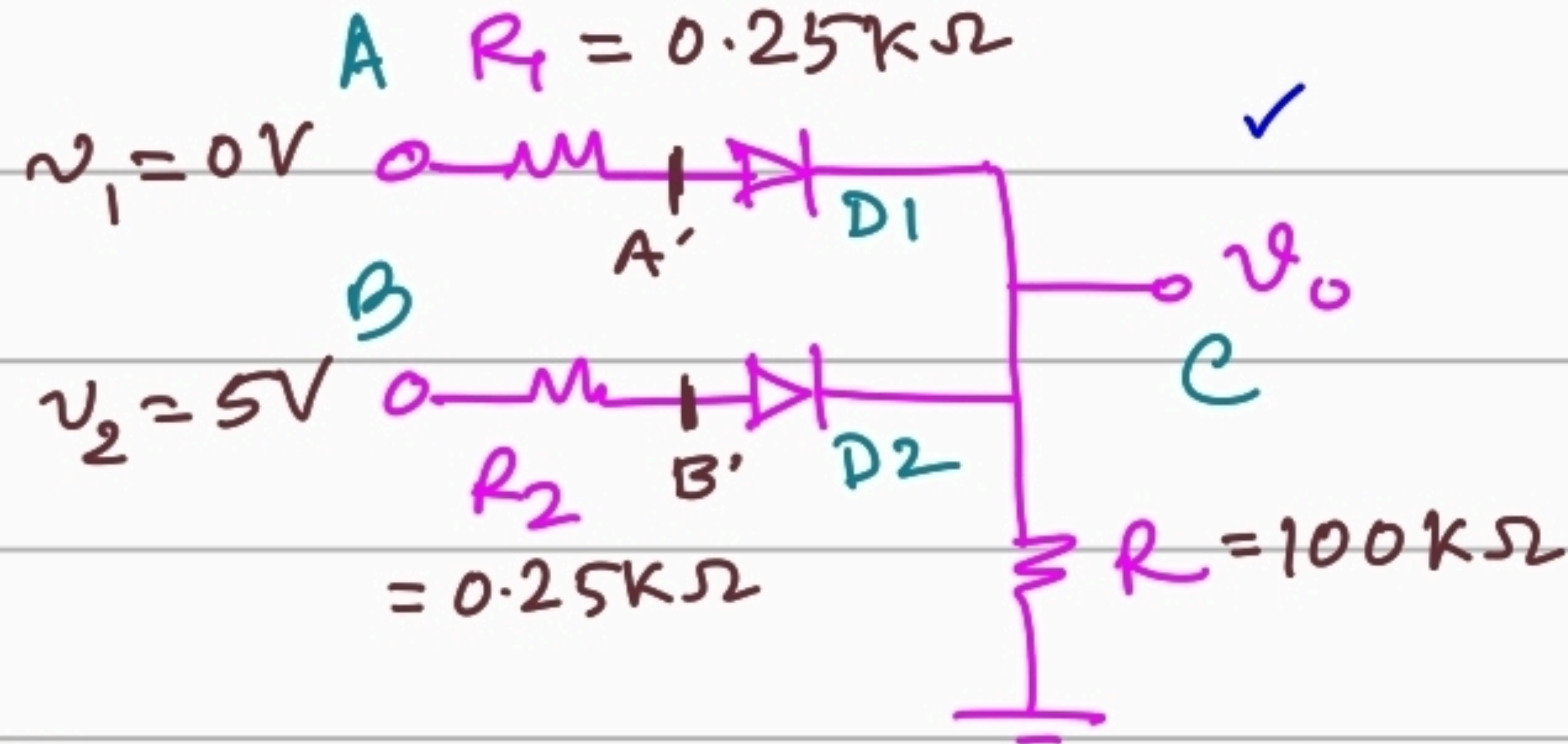




There is no higher voltage input terminal in this circuit. Thus diode will never turn on.

$$i = \frac{v_o - 0}{R} \Rightarrow 0 = \frac{v_o}{R} \quad \therefore \boxed{v_o = 0V}$$

Case ② & ③



const volt. drop model

Case ⓐ

* D1, OFF, D2, OFF

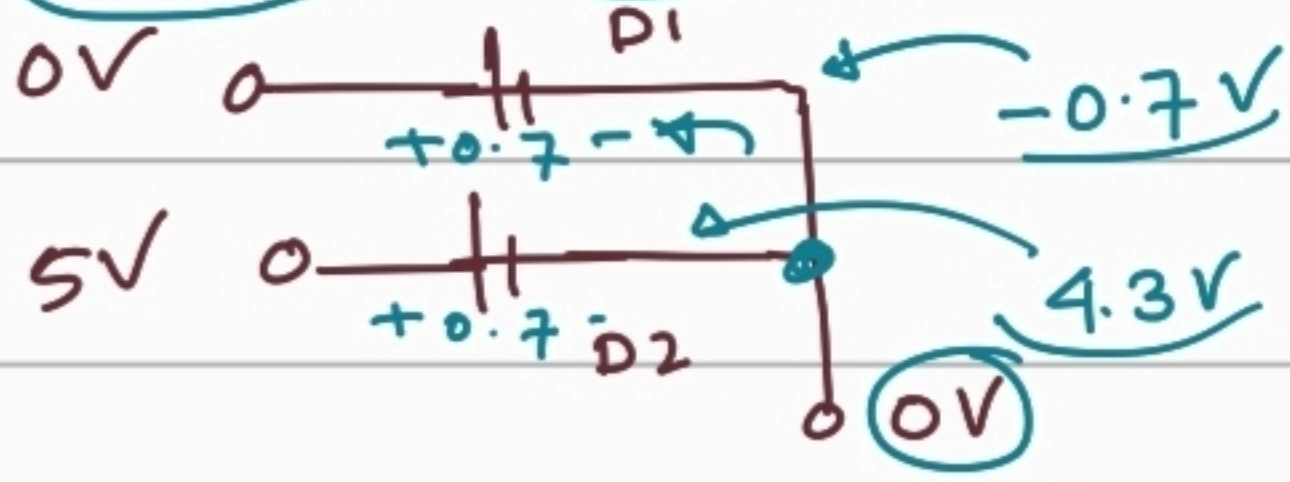
$v_{A'} = 0V$, $v_{B'} = 5V$

$v_0 = 0V$.

Contradiction.

* D1 ON, D2 OFF. # Contradiction.

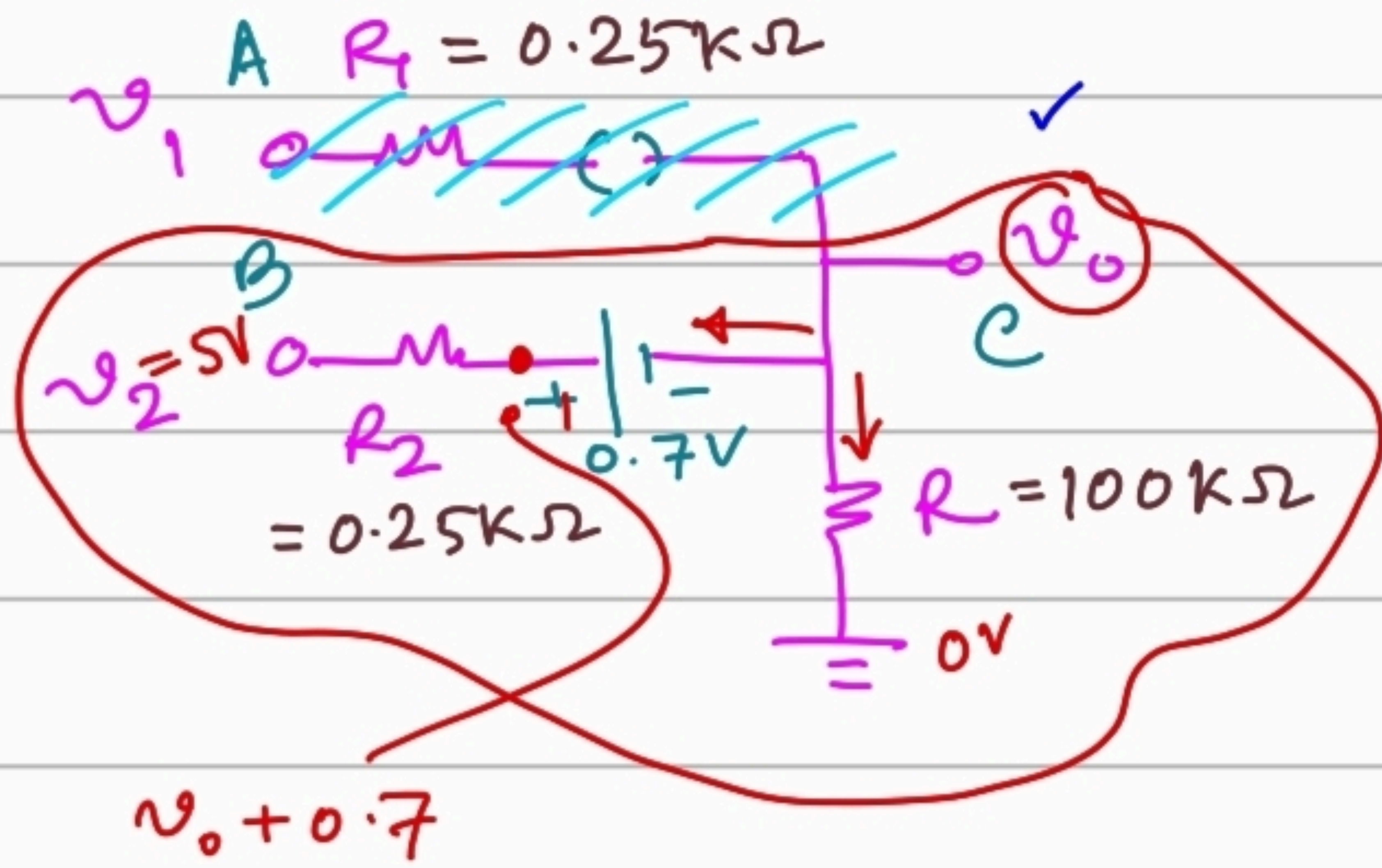
* D1 ON, D2, ON → Simplification



One node always has a single voltage.

If we do thorough analysis, we would find a contradictory result that current would flow in opposite direction through D1 diode.

* D1 OFF, D2 ON

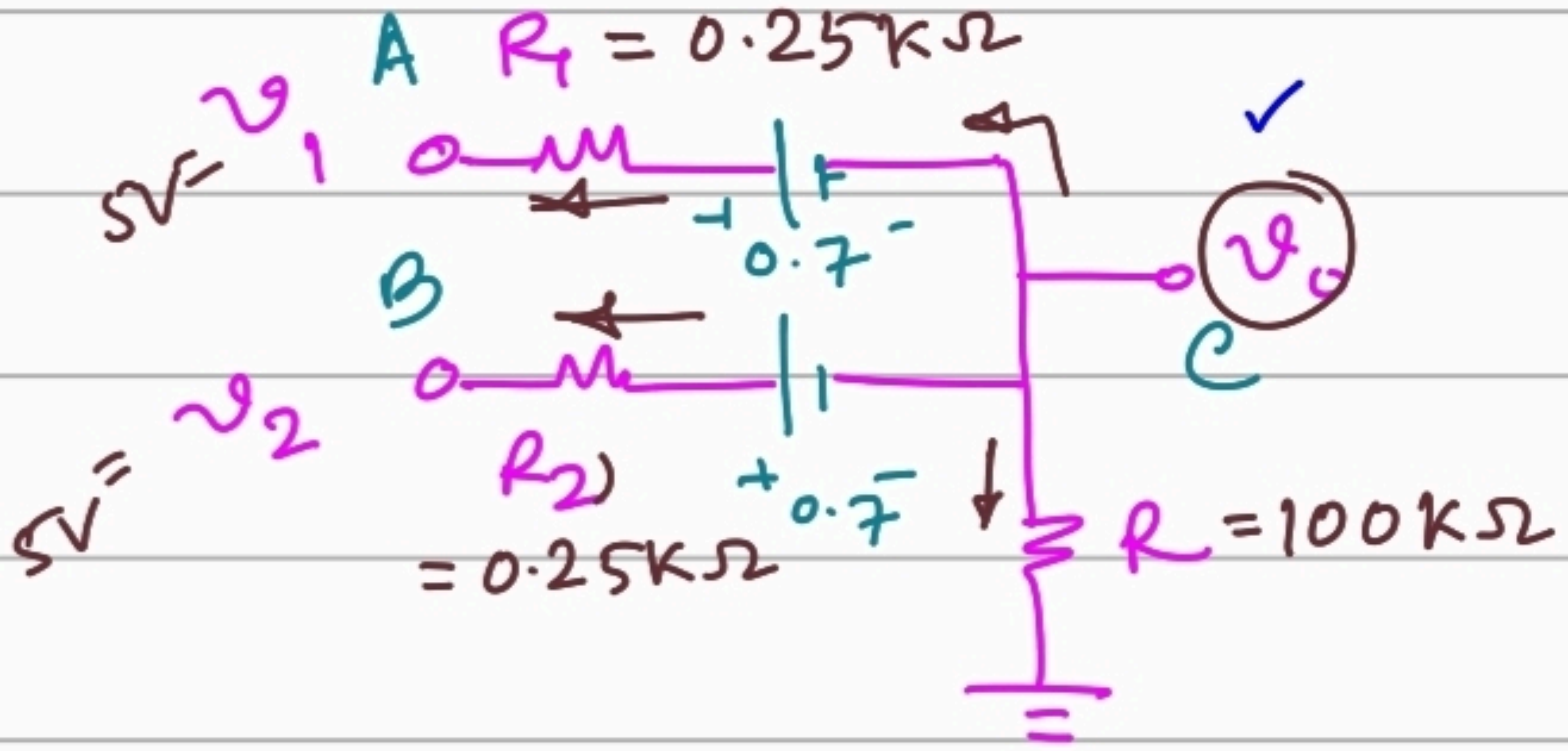


$$\frac{v_0 - 0}{100K} + \frac{(v_0 + 0.7) - 5V}{0.25K} = 0$$

$$\Rightarrow \boxed{v_0 = 4.2893\ V}$$

Case 4: $v_1 = 5V$, $v_2 = 5V$

* D_1 ON, D_2 ON.



$$\frac{v_o - 0}{100K} + \frac{(v_o + 0.7) - 5V}{0.25K} \times 2 = 0$$

$$v_o = 4.2946V$$

\# $\boxed{4.289,\ 4.294}$ We will choose the smallest of the high voltage as output voltage high.

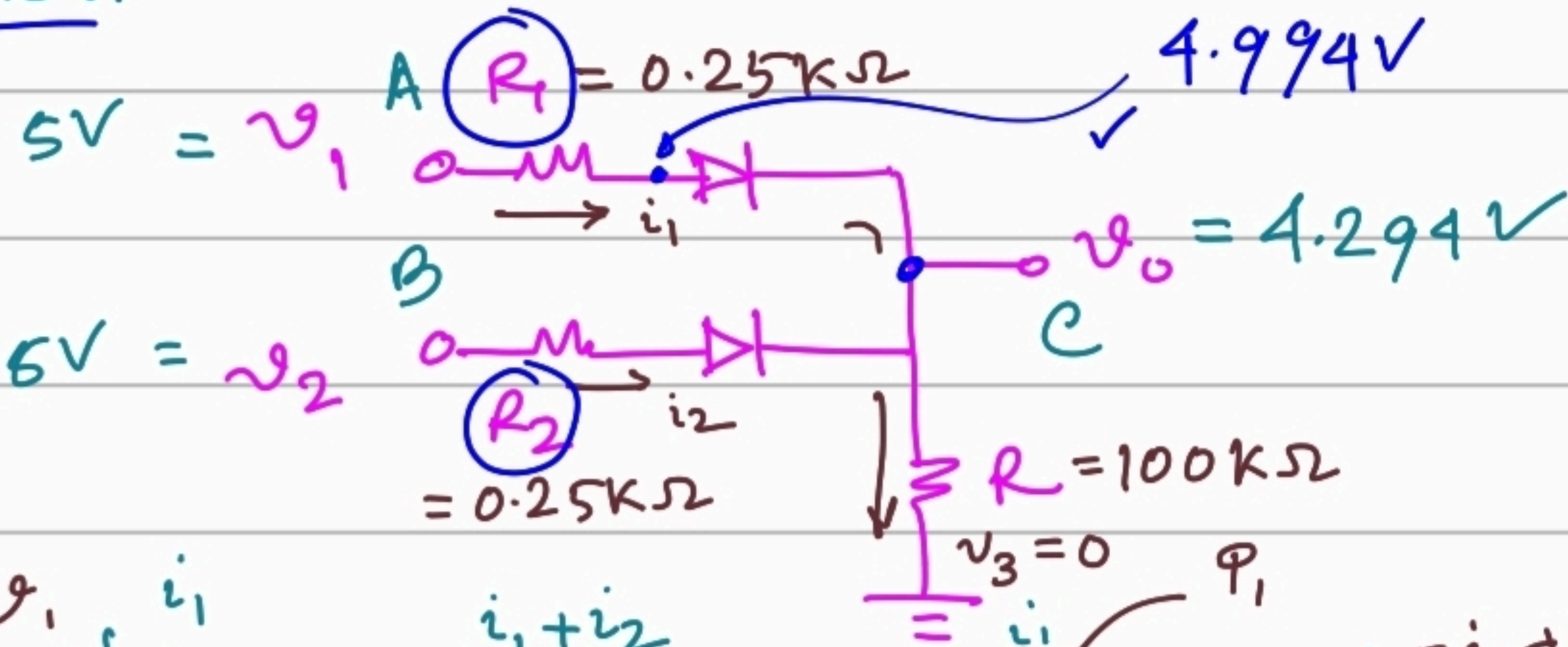
$$V_{OH} = 4.289\,V$$

Power Dissipation :- Maximum dissipation of power among four cases.

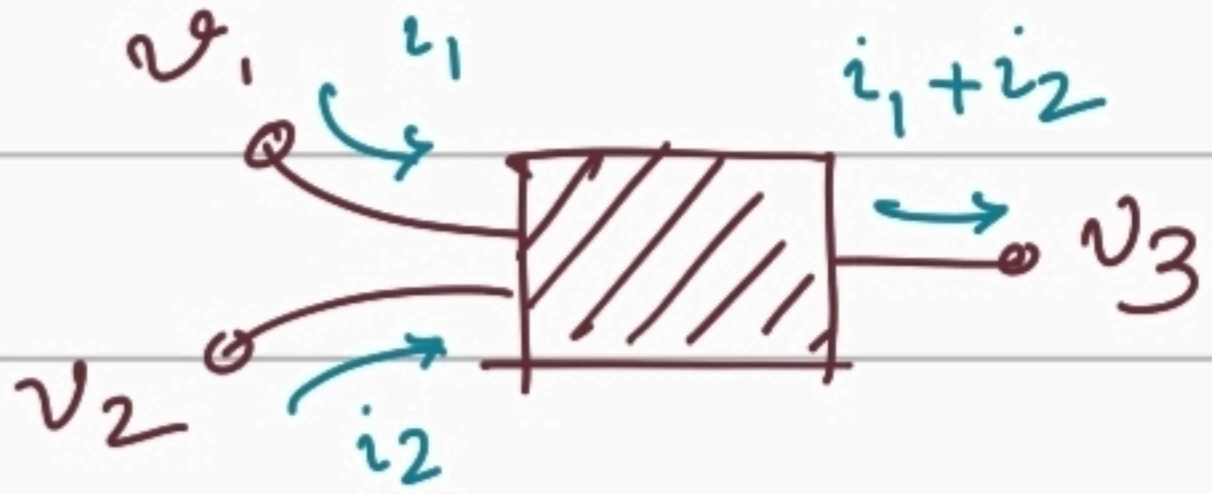
\# Case 4: Each individual branch is conducting current. Thus it has maximum power dissipation.

$$i_1 = \frac{5 - 4.994}{0.25K} = 0.0215\,mA$$

$$i_1 = i_2$$



Theory:



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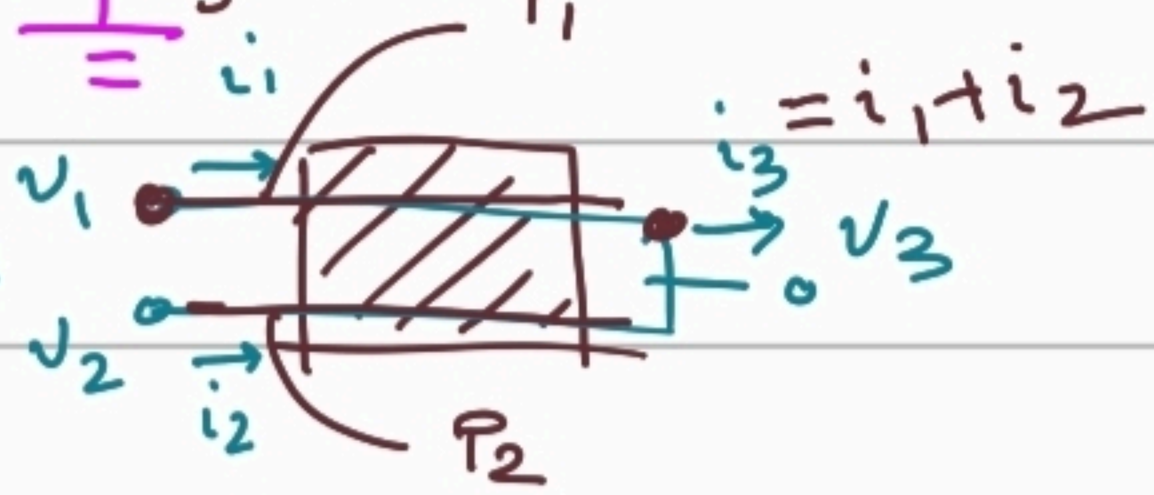
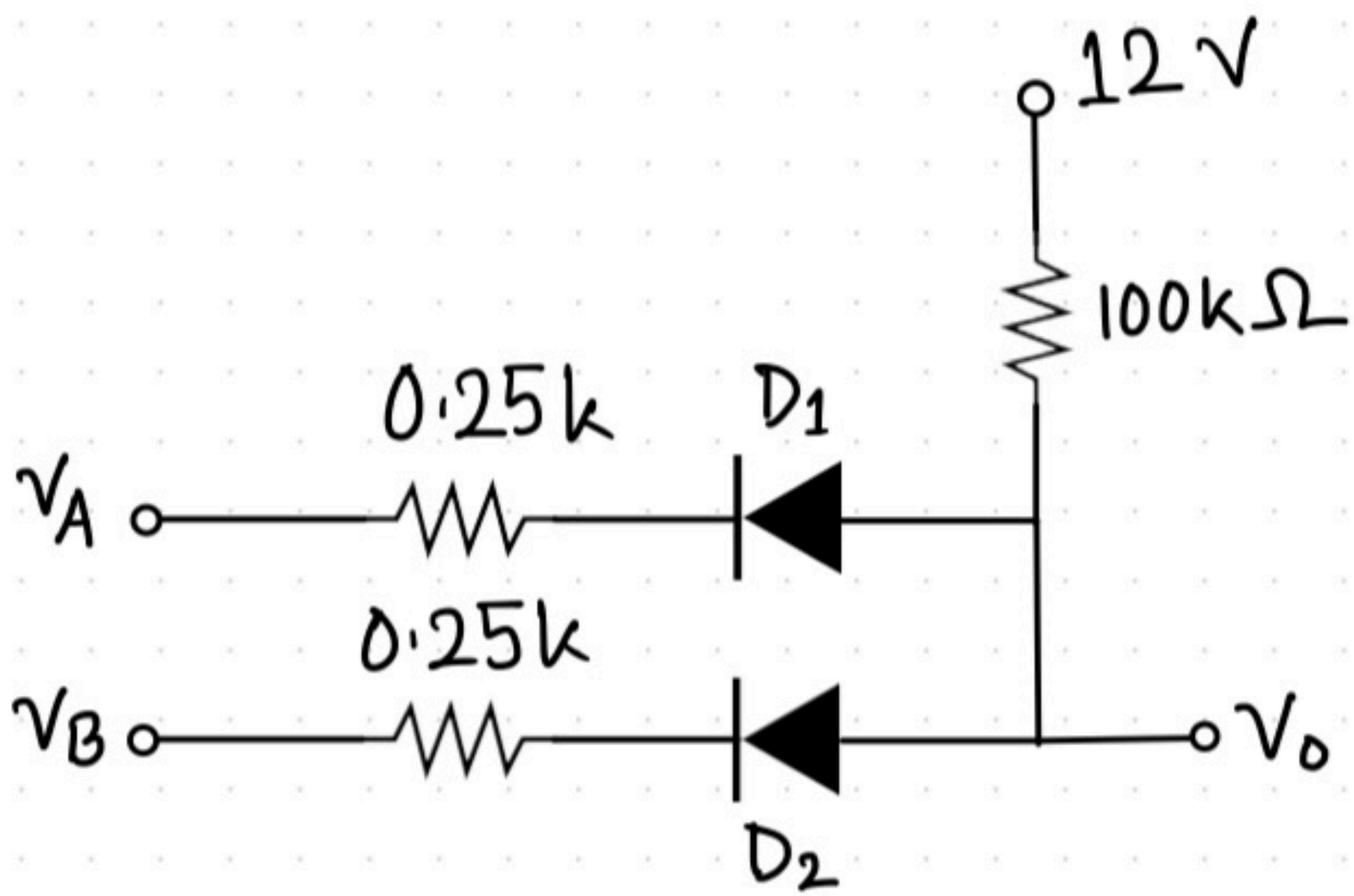


$$P_1 = (v_1 - v_3)\,i_1,\quad P_2 = (v_2 - v_3)\,i_2$$

Total power, $P = P_1 + P_2$

$$P_2 = P_1 = (5 - 0)V(0.0215\,mA) \Rightarrow P = 0.215\,mW$$

$$V \cdot \mu A = \mu W$$



Find out the output voltage levels for all input combinations.

Calculate the maximum power dissipation.

Solution:

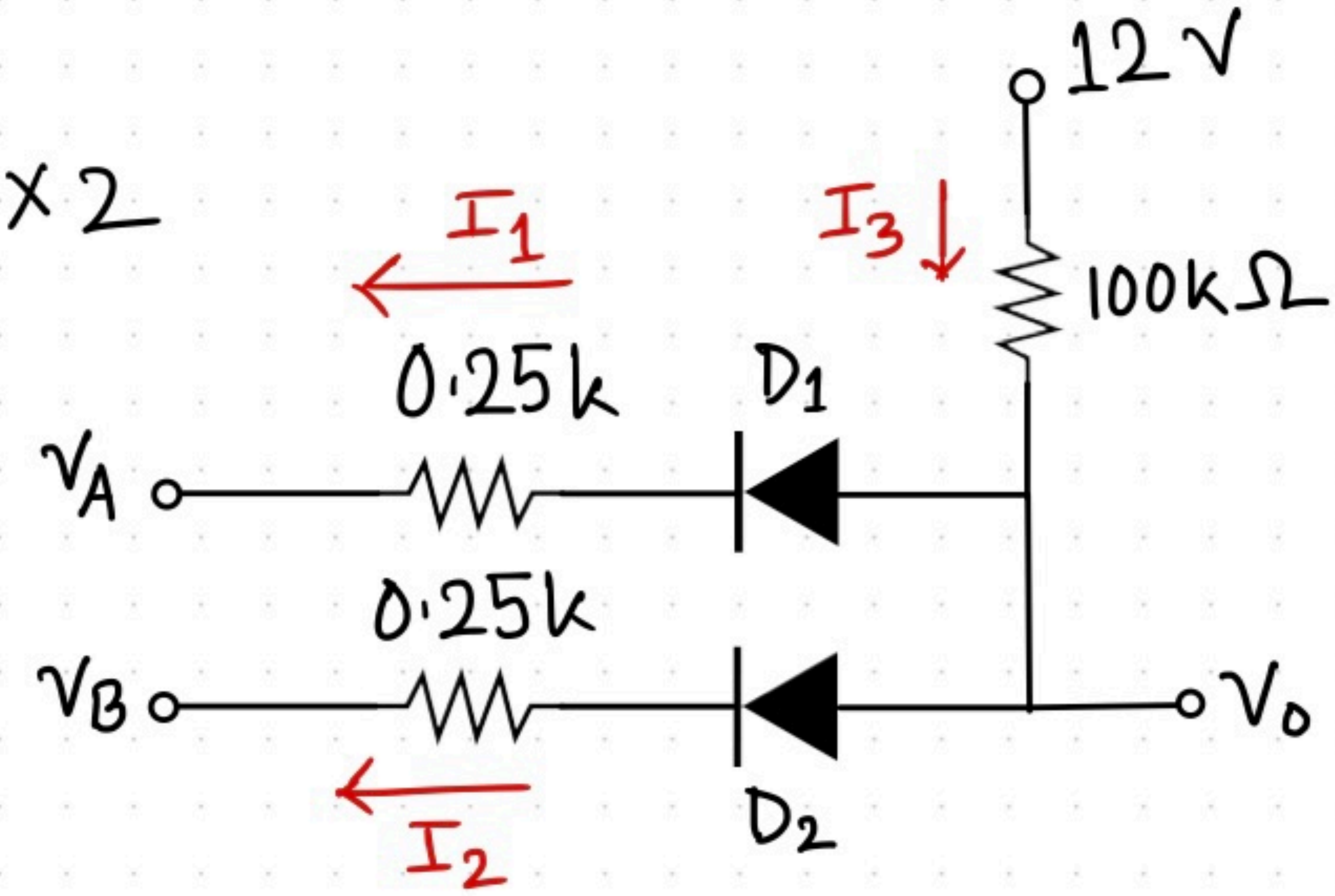
Case 1: For $V_A = 0\,V$, $V_B = 0\,V$

Both diodes D_1 and D_2 will conduct.

$\therefore I_3 = I_1 + I_2$

$$\Rightarrow \frac{12 - V_0}{100k} = \frac{(V_0 - 0.7) - 0}{0.25k} \times 2$$

$$\Rightarrow V_0 = 0.714\ V$$



Case 2 and 3 are identical

When any one input V_A or V_B is 0 V,

Any one diode will conduct.

$\therefore I_3 = I_1 + I_2$

$$\Rightarrow \frac{12 - V_0}{100k} = \frac{(V_0 - 0.7) - 0}{0.25k}$$

$$\Rightarrow V_0 = 0.728\ V$$

Case 4: For $V_A = 12\ V$ and $V_B = 12\ V$

Both diodes will be OFF.

$\therefore I_3 = 0,\ I_2 = 0,\ I_1 = 0$

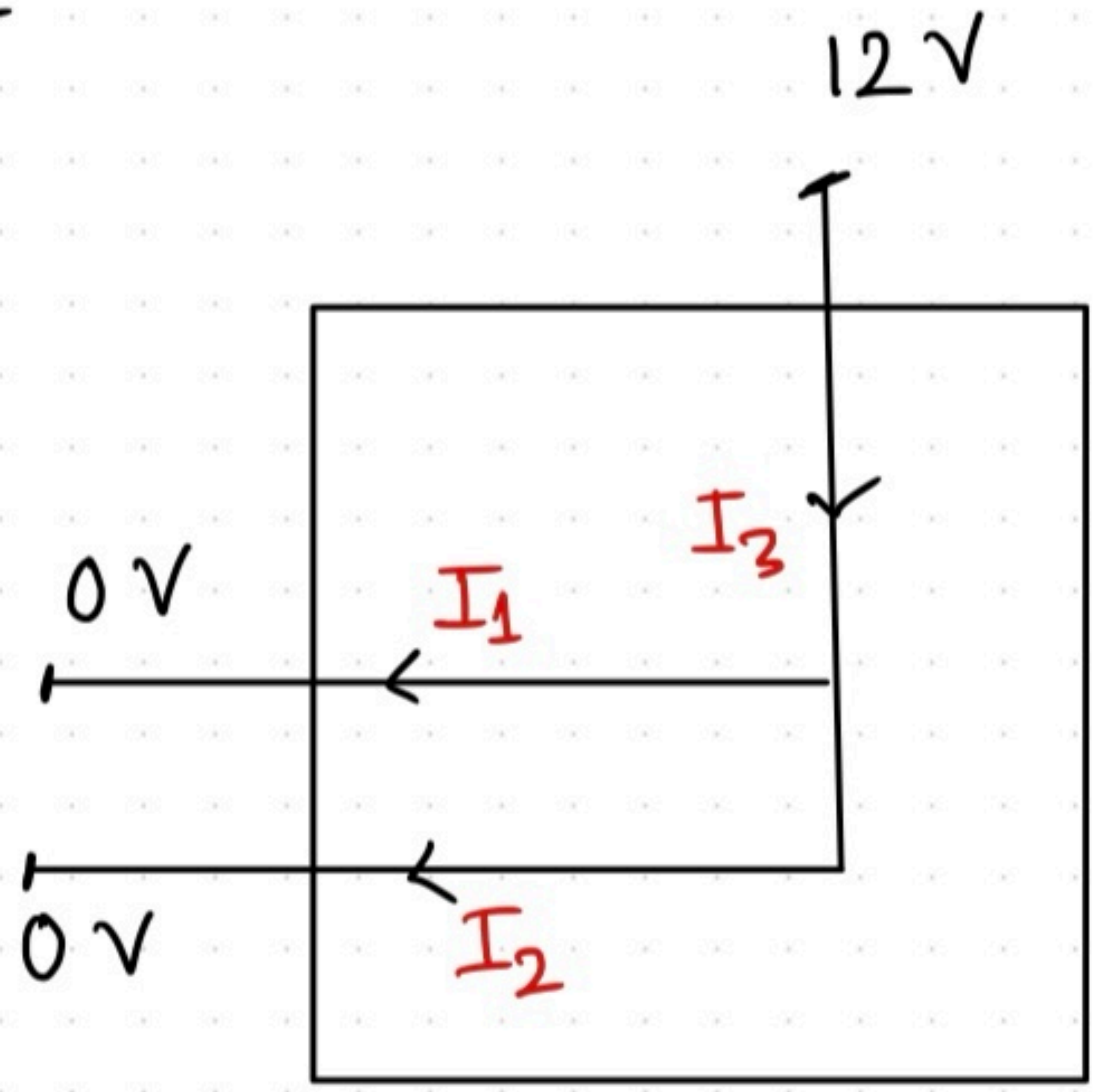
$\therefore V_o = 12\ V$

Maximum power dissipation happens when current is maximum i.e. case 1

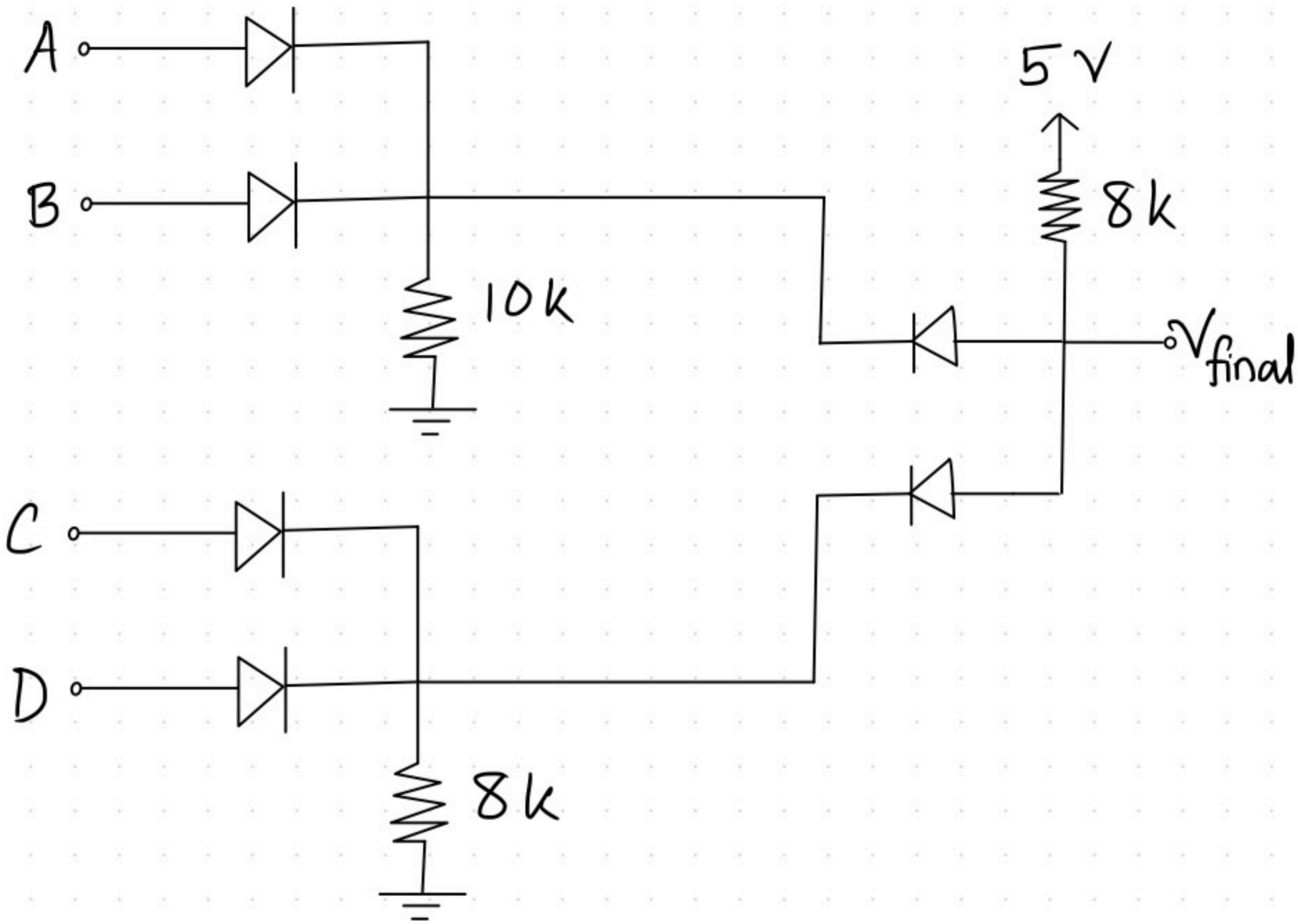
$P = \Delta V I$

$= (12 - 0) \times \dfrac{12 - 0.714}{100k}$

$= 1.354\ mW$



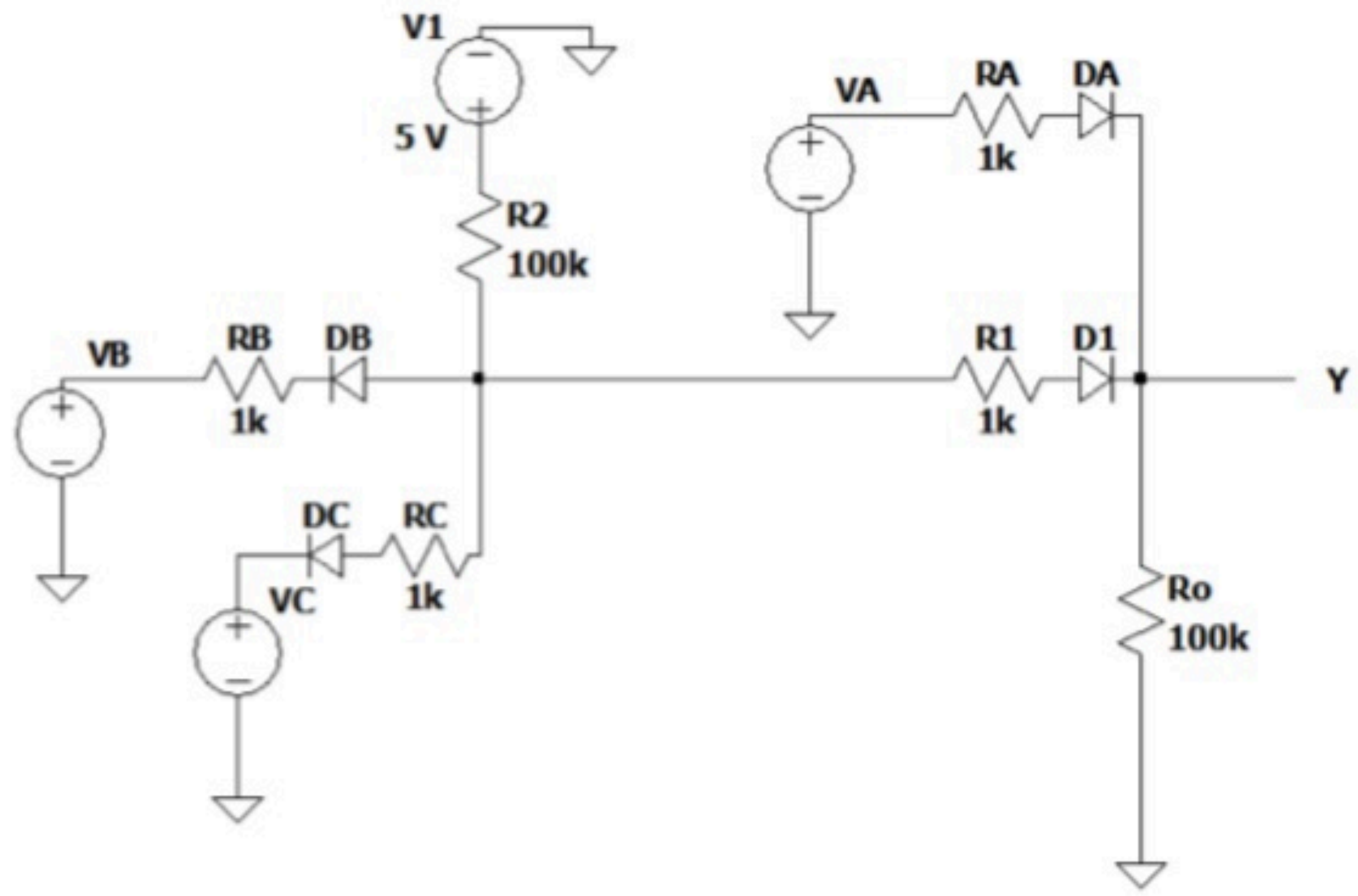
Question 2



Determine the boolean expression of V_{final} in terms of input A, B, C, D

Solution: V_{final} = (A+B).(C+D)

Exercise 4



For the diode logic circuit given-

a) Find the Boolean expression of Y.
b) Determine the higher & lower threshold of output voltage.
c) Find the maximum & average power dissipation of the full circuit.

Ans:

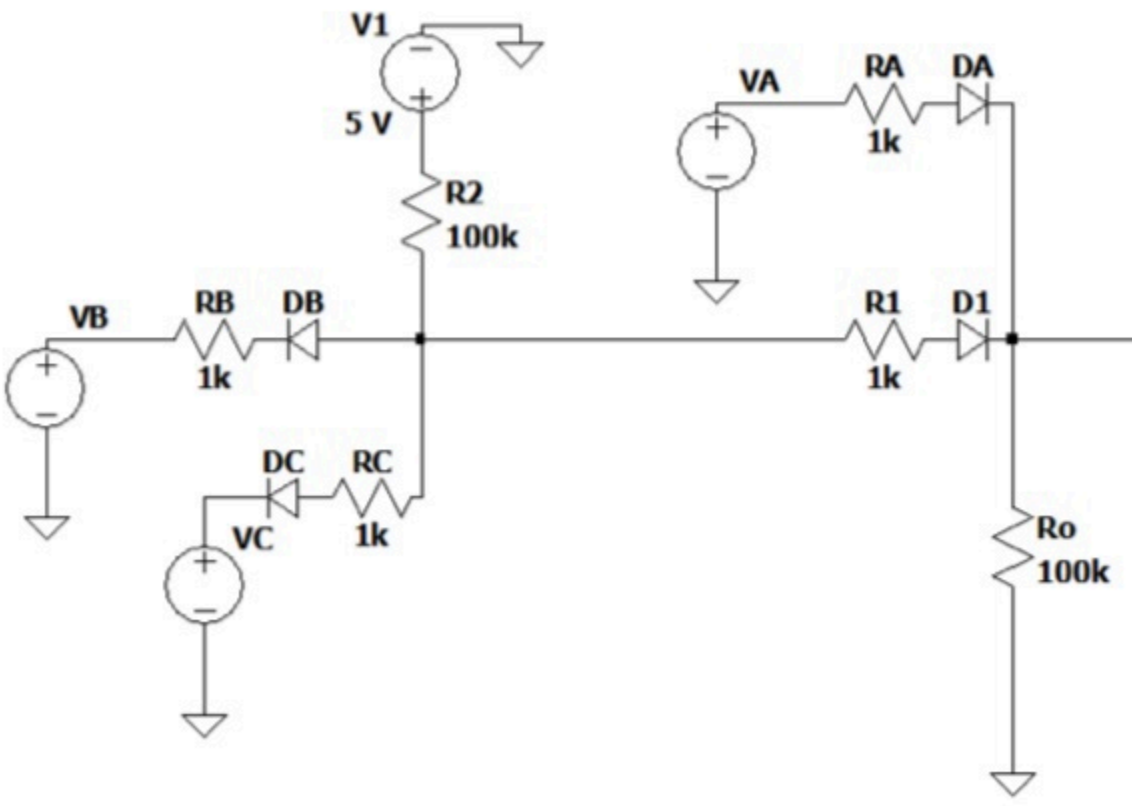
a) $Y = A + BC$

b) $2.14\ V, 0\ V$

c) $3.8\ mW, 0.742\ mW$

Exercise 4

Solution:



a) Y = A + BC

A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

b) Output voltage is high for the last 5 rows of input in the truth table.

Case (0,1,1):

$D_A, D_B \;\&\; D_C$ *all are off*, D_1 *on*

Hence, KCL at the output voltage node of the AND gate, $V_{o(and)}$ gives,

$$I_{R2} = I_{DB} + I_{DC} + I_{R1} \rightarrow \frac{5 - V_{o(and)}}{100} = 0 + 0 + \frac{V_{o(and)} - V_Y - 0.7}{1}$$

$$\rightarrow 101\, V_{o(and)} - 100 V_Y = 75$$

KCL at Y gives,

$$I_{R1} + I_A = I_{R_o} \rightarrow \frac{V_{o(and)} - V_Y - 0.7}{1} = \frac{V_Y - 0}{100} \rightarrow 100 V_{o(and)} - 101 V_Y = 70$$

Solving the two equations, $V_{o(and)} = 2.86\, V, V_Y = 2.14\, V$

Case (1,0,0):

$D_A, D_B \& D_C$ *all are on*, D_1 *can be assumed off (since output of AND gate is low)*

KCL at AND gate output node,

$$I_{R2} = I_{DB} + I_{DC} \rightarrow \frac{5 - V_{o(and)}}{100} = \frac{V_{o(and)} - 0 - 0.7}{1} \times 2 \rightarrow V_{o(and)} = 0.721\, V$$

KCL at Y,

$$I_{RA} = I_{R_o} \rightarrow \frac{5 - V_Y}{1} = \frac{V_Y - 0}{100} \rightarrow V_Y = 4.95\, V$$

Case (1,0,1):

D_A, D_B *on*, D_C *off*, D_1 *can be assumed off (since output of AND gate is low)*

KCL at AND gate output node,

$$I_{R2} = I_{DB} + I_{DC} \rightarrow \frac{5 - V_{o(and)}}{100} = \frac{V_{o(and)} - 0 - 0.7}{1} \rightarrow V_{o(and)} = 0.742\, V$$

KCL at Y,

$$I_{RA} = I_{R_o} \rightarrow \frac{5 - V_Y}{1} = \frac{V_Y - 0}{100} \rightarrow V_Y = 4.95\, V$$

Case (1,1,0): Same as Case (1,0,1)

Case (1,1,1):

D_A, D_1 *on, all other diodes are off*

KCL at Y,

$I_{RA} + I_{R1} = I_{R_o} \rightarrow \frac{5-V_Y-0.7}{1} + \frac{5-V_Y-0.7}{101} = \frac{V_Y-0}{100} \rightarrow V_Y = 4.25\,V$

$\therefore V_{o(and)} = 5 - I_{R1} \times 100 = 5 - \frac{5-4.94-0.7}{101} \times 100 = 4.96\,V$

Thus, the higher threshold of output voltage =
$\min(V_Y\ of\ all\ high\ output\ cases) = 2.14\,V$

Output voltage is low for the first 3 rows of input in the truth table.

Case (0,0,0):

$D_A\ off, D_B \& D_C\ on, D_1 can\ be\ assumed\ off$

(since the output of thte AND gate is low)

KCL at AND gate output node,

$I_{R2} = I_{DB} + I_{DC} \rightarrow \frac{5-V_{o(and)}}{100} = \frac{V_{o(and)}-0-0.7}{1} \times 2 \rightarrow V_{o(and)} = 0.721\,V$

$V_Y = 0$ *(since all diode branches connected to Y are open)*

Case (0,0,1):

$D_A\ off, D_B\ on, D_C\ off, D_1\ can\ be\ assumed\ off$

KCL at AND gate output node,

$I_{R2} = I_{DB} + I_{DC} \rightarrow \frac{5-V_{o(and)}}{100} = \frac{V_{o(and)}-0-0.7}{1} \rightarrow V_{o(and)} = 0.742\,V$

$V_Y = 0\ (for\ the\ same\ reason\ as\ previous\ case)$

Case (0,1,0): Same as Case (0,0,1)

Thus, lower threshold of output voltage $= \max(V_Y\ of\ all\ low\ output\ cases)$
$= 0\ V$

c) Power dissipation of all cases:

Case (0,0,0):

$$P_{0,0,0} = (5-0) \times I_{R2} = 5 \times \frac{5-0.721}{100} = 0.214\ mW$$

Case (0,0,1):

$$P_{0,0,1} = (5-0) \times I_{R2} = 5 \times \frac{5-0.742}{100} = 0.213\ mW$$

Case (0,1,0): Same as Case (0,0,1).

Case (0,1,1):

$$P_{0,1,1} = (5-0) \times I_{R2} = 5 \times \frac{5-2.86}{100} = 0.107\ mW$$

Case (1,0,0):

$$P_{1,0,0} = (5-0) \times I_{R2} + (5-0) \times I_{RA} = 5 \times \left(\frac{5-0.721}{100} + \frac{5-4.95}{1}\right) = 0.464\ mW$$

Case (1,0,1):

$$P_{1,0,1} = (5-0) \times I_{R2} + (5-0) \times I_{RA} = 5 \times \left(\frac{5-0.742}{100} + \frac{5-4.95}{1}\right) = 0.463\ mW$$

Case (1,1,0): Same as Case (1,1,0).

Case (1,1,1):

$$P_{1,1,1} = (5-0) \times I_{R1} + (5-0) \times I_{RA} = 5 \times \left(\frac{4.96-4.25-0.7}{1} + \frac{5-4.25}{1}\right) = 3.8\ mW$$

$\therefore\ Maximum\ Power\ Dissipation, P_{max} = 3.8\ mW$

$\therefore\ Average\ Power\ Dissipation, P_{avg} = \frac{0.214+0.213\times2+0.107+0.464+0.463\times2+3.8}{8}$

$= 0.742\ mW$